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Education

- CS + Stats University of British Columbia (2029 expected), Vancouver. GPA: 93.4%
- Coursework: Systematic Program design (Racket), Data science in R, Discrete Math, OOP Java, Design Patterns, Software development Java
- Certifications: Deep Learning Specialization by Andrew Ng [[Verify Certificate](#)]

Awards

- Karen McKellin International Leader of Tomorrow scholar (Full ride, 1 of 30 worldwide).
- Dean's scholar.

Skills

Languages: Python, JavaScript, Java

Frameworks: React, FastAPI, LangChain, LangGraph, JUnit, Pytest, Pytorch, Supabase, Pinecone, ChromaDB ,Django

AI/ML: RAG pipelines, agentic AI & multi-step agent architectures, ReAct, RAGAS evaluation

Projects

GlimpseUganda [[Demo](#)][[Live app](#)] [[Code](#)] React, FastAPI, LangChain, Supabase, Pinecone, Pytest, CICD, RAG **Apr 2026**

- Developed an interactive map based chatbot where users ask touristic sites questions to improve tourism accessibility. Submitted to the Government Systems Prototype Showcase, Ministry of ICT Uganda.
- Engineered ingestion pipeline: documents are LLM validated for relevance, uploaded to Supabase, embedded and synced to Pinecone vector db via webhooks.
- Achieved 90% context recall measured via RAGAS evaluation by implementing reranking for retrieval quality.
- Automated testing and deployment using GitHub actions, deploying web app to Vercel and Railway.

StreetWatch [[Demo](#)] [[Live app](#)] [[Code](#)] React, FastAPI, Supabase, Pytorch, REST APIs **Apr 2026**

- Engineered a road surveillance app where road users upload pictures of road hazards through their mobile phone reducing surveillance costs for authorities.
- Integrated EfficientNet machine learning model to verify user uploaded photos.
- Implemented client-side image compression (5MB -> 400KB) for low bandwidth access.
- Implemented rate limiting (10 reqs/minute) and PostGIS spatial deduplication.
- Deployed app using Vercel and Railway.

Rubis data entry agent [[Code](#)] LangGraph, Tool calling, Agentic AI, ReAct, React, FastAPI **Jun 2026**

- Built an agentic data entry system using LangGraph that parses natural language and populates excel spreadsheets with reasoning loops and tool calling.
- Wrote test suite to validate agentic behavior.

UCF crime dataset Computer vision & ML project [[Code](#)] Pytorch, Scikit-learn, NumPy, Pandas **Jun 2025**

- Fine-tuned Facebook's SlowFast CNN to classify 11 types of criminal activity from CCTV footage.
- Used a HuggingFace sentiment model to score violence from JSON annotations for targeted clip extraction.

Rubis Energies dashboard app [[Code](#)] **Jul 2023 – Dec 2023**

- Engineered a dashboard app to generate reports using React and Django replacing manual spreadsheet workflows.

Experience

Reinforcement Learning research UBC Undergraduate Research Organization [[Code](#)] **Sept 2025 – Feb 2026**

- Selected as part of a mentorship program to conduct research on sampling experiences in experience replay.
- Designed a custom replay buffer that prioritizes experiences by learning gain (loss before vs after training) an alternative to temporal difference prioritization under the guidance of a senior mentor.

ICT Operations Intern, United Nations High Commission for Refugees Uganda **Jul 2024 – Aug 2024**

- Resolved hardware and software faults across 40-50+ devices a day including system setup and configuration.
- Managed refugee call center operations, web-conference set up and LAN cabling infrastructure.

Hardware intern, Millenium Infosys Uganda **May 2022 – Jun 2022**

- Repaired and configured 80+ laptops and desktops daily including hardware upgrades and troubleshooting while managing client-side technical complaints.

Leadership Experiences: Technical director UBC Bangladeshi Students' Association 2025/26 & VP Tech 2026 - present